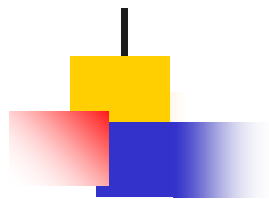


Diffusion Models in CV

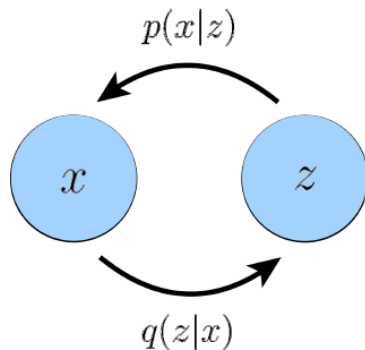


Amit K Roy-Chowdhury

Acknowledgments: <https://calvinyluo.com/2022/08/26/diffusion-tutorial.html#evidence-lower-bound>
<https://arxiv.org/abs/2403.18103>

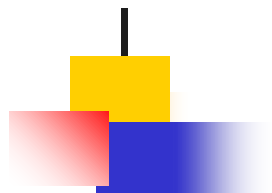


Standard VAE



$$\ln p(\mathbf{x}) \geq \underbrace{\mathbb{E}_{\mathbf{z} \sim q_{\phi}(\mathbf{z}|\mathbf{x})} [\ln p(\mathbf{x}|\mathbf{z})]}_{\text{Reconstruction Error}} - \underbrace{\mathbb{E}_{\mathbf{z} \sim q_{\phi}(\mathbf{z}|\mathbf{x})} [\ln q_{\phi}(\mathbf{z}|\mathbf{x}) - \ln p(\mathbf{z})]}_{\text{KL Divergence}}$$

Objective: $\arg \max_{\phi, \theta} \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} [\log p_{\theta}(\mathbf{x} | \mathbf{z})] - \mathcal{D}_{\text{KL}}(q_{\phi}(\mathbf{z} | \mathbf{x}) || p(\mathbf{z}))$

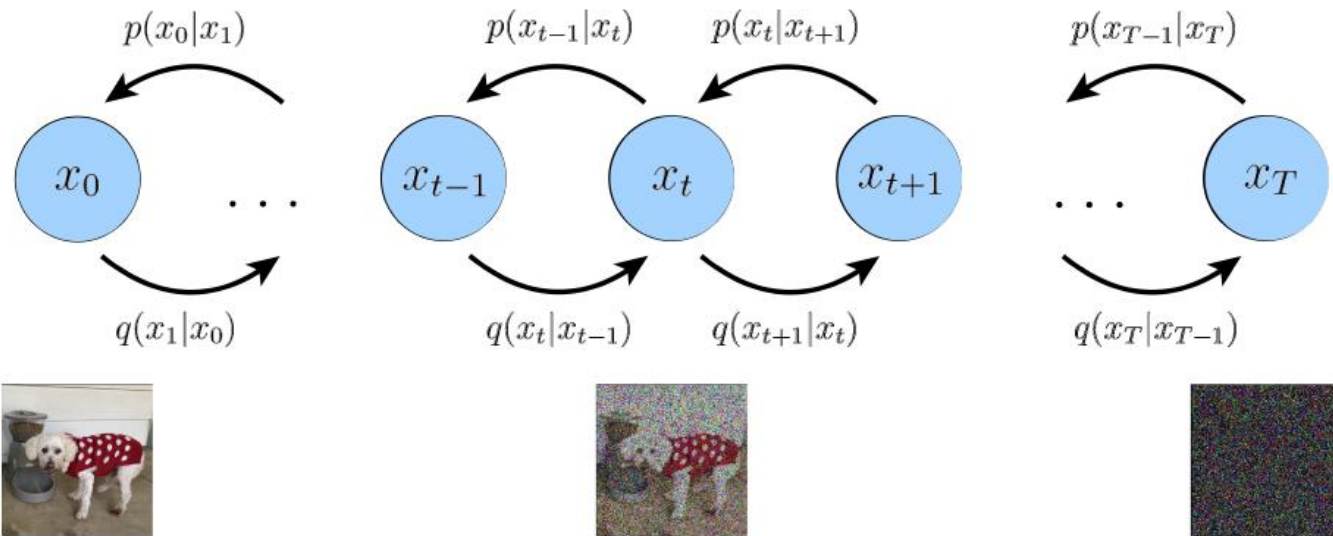


Markov Chain

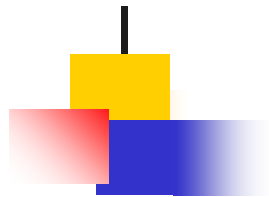
A discrete-time Markov chain $\{X_n | n = 0, 1, \dots\}$ is a discrete-time, discrete-value random sequence such that given $X_0, \dots, X_n$, the next random variable X_{n+1} depends only on X_n through the transition probability

$$P[X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0] = P[X_{n+1} = j | X_n = i] = P_{ij}.$$

Variational Diffusion Models



- Learn $p_{\theta}(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$
- Sample Gaussian noise from $p(\mathbf{x}_T)$ and denoise based on $p_{\theta}(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ for T steps



Variational Diffusion Models (VDMs)

Latent dimension is the same as data dimension: both data and latent variables as $\mathbf{x}_t$

$$q(\mathbf{x}_{1:T} \mid \mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t \mid \mathbf{x}_{t-1})$$

Structure of the latent encoder at each step is a linear Gaussian model

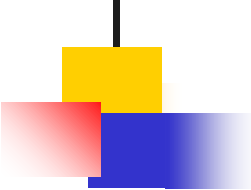
$$q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t} \mathbf{x}_{t-1}, (1 - \alpha_t) \mathbf{I})$$

α_t evolves such that the latent at the final timestep is a standard Gaussian.

$$p(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^T p_{\theta}(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$$

where,

$$p(\mathbf{x}_T) = \mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I})$$



Variational Diffusion Models (VDMs)

Denoising Diffusion Probabilistic Models (DDPM)

Because calculating the true exact posterior $q(x_{t-1}|x_t)$ is mathematically intractable (it requires knowing the entire data distribution), we use a neural network to parameterize and approximate a Gaussian distribution.

The formal mathematical expression for the reverse denoising process is:

$$p_{\theta}(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_{\theta}(x_t, t), \Sigma_{\theta}(x_t, t))$$

During implementation, instead of predicting μ_{θ} entirely from scratch, the neural network is trained to predict the noise $\epsilon_{\theta}(x_t, t)$ that was injected at that step. The mean is then reconstructed using the network's noise prediction via this formula:

$$\mu_{\theta}(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_{\theta}(x_t, t) \right)$$

Where $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$ represents the cumulative noise schedule up to step t .

- $p_{\theta}(x_{t-1}|x_t)$: The learned probability distribution of the cleaner image x_{t-1} given the noisier image x_t .
- $\mu_{\theta}(x_t, t)$: The **mean** of the distribution, predicted by your neural network (typically a U-Net or a Diffusion Transformer). It represents the model's best guess of what the slightly denoised image features look like at timestep t .
- $\Sigma_{\theta}(x_t, t)$: The **variance** of the distribution. In the foundational DDPM paper, this is often treated as a fixed, non-learned time-dependent constant ($\sigma_t^2 \mathbf{I}$) proportional to the noise schedule, though modern variants sometimes predict it to improve sample speed and quality.

Denoising Diffusion Restoration Models (DDRM)

The core mathematical breakthrough of DDRM is how it handles the corruption. If your degraded image y is modeled as:

$$y = Hx + z$$

(Where H is the degradation operator like blurring or masking, x is the clean image, and z is noise).

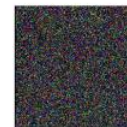
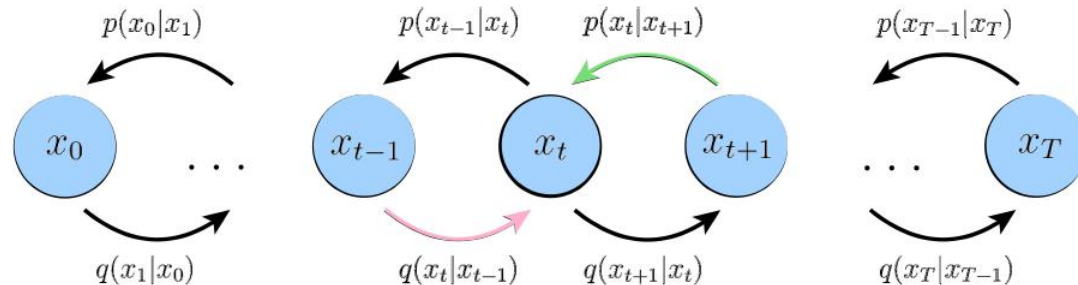
- **For uncorrupted/accessible frequencies** ($\sigma_i > 0$): It takes the known data from y and denoises it slightly, regularizing it with the model's prediction ϵ_{θ} .
- **For completely lost frequencies** ($\sigma_i = 0$, like missing pixels in a mask): It relies 100% on the pre-trained diffusion model's prediction to "hallucinate" and generate the missing details from scratch.

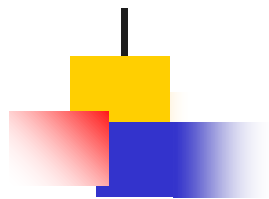


Additional Reading

ELBO for VDMs

$$\begin{aligned}
 \log p(\mathbf{x}) &\geq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T} | \mathbf{x}_0)} \right] \\
 &= \underbrace{\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_0 | \mathbf{x}_1)]}_{\text{reconstruction term}} - \underbrace{\mathbb{E}_{q(\mathbf{x}_{T-1}|\mathbf{x}_0)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_T | \mathbf{x}_{T-1}) || p(\mathbf{x}_T))]}_{\text{prior matching term}} \\
 &\quad - \underbrace{\sum_{t=1}^{T-1} \mathbb{E}_{q(\mathbf{x}_{t-1}, \mathbf{x}_{t+1}|\mathbf{x}_0)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_t | \mathbf{x}_{t-1}) || p_\theta(\mathbf{x}_t | \mathbf{x}_{t+1}))]}_{\text{consistency term}}
 \end{aligned}$$





VDM Distribution

$$\mathbf{x}_t \sim q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) \quad q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t} \mathbf{x}_{t-1}, (1 - \alpha_t) \mathbf{I})$$

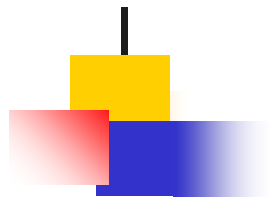
$$\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_{t-1} + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon} \quad \text{with } \boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{\epsilon}; \mathbf{0}, \mathbf{I})$$

$$\mathbf{x}_{t-1} = \sqrt{\alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{1 - \alpha_{t-1}} \boldsymbol{\epsilon} \quad \text{with } \boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{\epsilon}; \mathbf{0}, \mathbf{I})$$

Conditional Distribution:

$$q_{\phi}(\mathbf{x}_t \mid \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t \mid \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}),$$

$$\text{where } \bar{\alpha}_t = \prod_{i=1}^t \alpha_i.$$



VDM Distribution - Proof

$$\begin{aligned}
 \mathbf{x}_t &= \sqrt{\alpha_t} \mathbf{x}_{t-1} + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_{t-1} \\
 &= \sqrt{\alpha_t} (\sqrt{\alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{1 - \alpha_{t-1}} \boldsymbol{\epsilon}_{t-2}) + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_{t-1} \\
 &= \underbrace{\sqrt{\alpha_t \alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{\alpha_t} \sqrt{1 - \alpha_{t-1}} \boldsymbol{\epsilon}_{t-2}}_{\mathbf{w}_1} + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_{t-1}.
 \end{aligned}$$

Sum of two Gaussians is a Gaussian

Mean is zero

Covariance: $\mathbb{E}[\mathbf{w}_1 \mathbf{w}_1^T] = [(\sqrt{\alpha_t} \sqrt{1 - \alpha_{t-1}})^2 + (\sqrt{1 - \alpha_t})^2] \mathbf{I}$
 $= [\alpha_t(1 - \alpha_{t-1}) + 1 - \alpha_t] \mathbf{I} = [1 - \alpha_t \alpha_{t-1}] \mathbf{I}$

Thus:

$$\begin{aligned}
 \mathbf{x}_t &= \sqrt{\alpha_t \alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{1 - \alpha_t \alpha_{t-1}} \boldsymbol{\epsilon}_{t-2} \\
 &= \sqrt{\alpha_t \alpha_{t-1} \alpha_{t-2}} \mathbf{x}_{t-3} + \sqrt{1 - \alpha_t \alpha_{t-1} \alpha_{t-2}} \boldsymbol{\epsilon}_{t-3} \\
 &= \vdots \\
 &= \left(\sqrt{\prod_{i=1}^t \alpha_i} \right) \mathbf{x}_0 + \left(\sqrt{1 - \prod_{i=1}^t \alpha_i} \right) \boldsymbol{\epsilon}_0.
 \end{aligned}$$

Define $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$

Then: $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}_0.$



$\mathbf{x}_t \sim q_{\phi}(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t | \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$